

Announcements

- EBIO party today

Today

- Agent-based model of death
 - higher scale outcome: number dead
- Binomial distribution as emergent process

Agent-based model

- Another outcome: **number dead** out of 100 individuals

Algorithm

set each individual to “alive”

set each individual to $p_death = 0.2$

for each individual

 generate U from $U(0,1)$

 if $U \leq p_death$

 set individual to “dead”

count dead

Binomial distribution

- **Emerges** from the agent-based simulation
- Underlying process: stochastic binary events
- Number of events that occur out of N possible
- **Higher-level abstraction** of the biological process or ABM is possible
 - i.e. simplifies to: $N_{\text{dead}} \sim \text{Binomial}(N, p)$

Binomial distribution

- Sum of independent stochastic binary events with the same probability

- $\Pr(X = x \mid n, p) =$

Discrete x
 $0, 1, 2, \dots, n$

$$f(x, n, p) = \binom{n}{x} p^x (1-p)^{n-x}$$

Probability
mass
function
(pmf)

$$X \sim \text{Binomial}(n, p)$$

Binomial coefficient:
how many ways to
choose x from n

“Probability that x events occur out of n possible”